GIRLS IN QUANTUM | QUANDELA

QashFlow

OPTION PRICING WITH QUANTUM MACHINE LEARNING

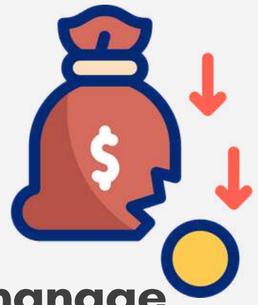
Qashflow | QHACKATHON

The Problem

Pricing Complex Financial Derivatives

Financial institutions must price swaptions accurately to:

01



manage financial risk

02

make informed trading decisions

03

ensure market stability

The Challenge

Limits of Traditional & AI Models

Even advanced deep learning models struggle with hundreds of correlated financial features on simplified assumptions that rarely hold in real markets.

Core Challenge

How can we accurately model complex financial instruments in a high-dimensional, non-linear market environment?

Why Quantum?Why Quandela?

Quantum Advantage for Feature Spaces as Quantum systems naturally operate in exponentially large spaces.

Quandela Photonic quantum circuits can:

- Capture complex nonlinear relationships
- Project data into richer feature spaces

Our Research Journey

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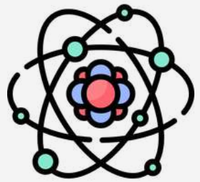
WE TESTED 9 DIFFERENT MODELS

FROM CLASSICAL → HYBRID → QUANTUM



Qashflow

Model Progression

Model	Modes	Photons	MSE	MAE	R2
Classical LSTM	—	—	0.34264	—	—
Hybrid Q-LSTM	6	2	0.20765	0.36580	—
Hybrid Q-Transformer	6	2	0.15485	0.32732	—
Hybrid Q-GRU	6	2	0.12054	0.26349	—
Hybrid Q-TCN	6	2	0.07043	0.20995	—
Hybrid Q-TCN	6	3	0.09997	0.24012	—
Hybrid Q-TCN	6	4	0.07775	0.22061	—
Hybrid Q-TCN	6	5	0.07482	0.21329	—
HPQRC	**8**	**3**	**1.93e-05**	**0.00389**	**0.998**

Our Final Solution

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HPQRC: HYBRID PHOTONIC-QUANTUM RESERVOIR COMPUTING

PIPELINE:

prices (224 features)

↓

PCA compression → 5 principal components

↓

Quantum reservoirs (parallel)

↓

Photonic time-delay reservoir (temporal memory)

↓

Ridge regression output

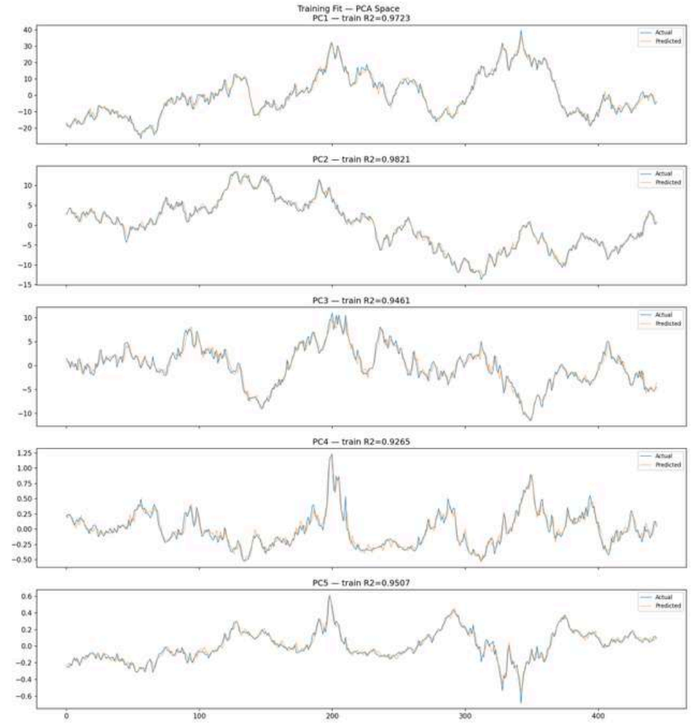

Figure 2: HPQRC fit in PCA space. R2: 0.97, 0.98, 0.95, 0.93, 0.95.

Qashflow

Let’s see the Website

Click Here

Results

Training data: 450 days

Final model performance:

- $R^2 = 0.998$
- MSE = 1.93×10^{-5}
- <2% prediction error on most days

Compared to classical models:

- Much higher accuracy
- Strong generalization

EVERY QUANTUM-ENHANCED MODEL **OUTPERFORMED** THE CLASSICAL BASELINE, DEMONSTRATING THE REAL ADVANTAGE OF QUANTUM FEATURE SPACES.

Conclusion

Quantum computing is not a magic layer.

But when the pipeline is designed correctly:

- Quantum circuits extract powerful features
- Hybrid models outperform classical baselines
- Financial prediction accuracy significantly improves

Qashflow demonstrates a practical path for Quantum ML in finance.

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Thank You

WHERE QUANTUM MEETS REAL-WORLD IMPACT

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